The Macroeconomic Effects of Lifting Barriers to Migration

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CRNYU

Stark Urban-Rural Gap in Low-Income Countries

Pictures show a rural Kenyan home and Nairobi, the capital city of Kenya. Source: Jocelyn Diaz (left) and Catalin Marin (right).

Returns to Migration Are High

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- urbanization remains limited
- growing randomized controlled trial evidence on why!

Why Do Gaps Persist and What Can We Do? RCT Evidence

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which interventions are most effective?

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which barriers are costliest?
which interventions are most effective?

- unclear because experiments have
  - different environment and sampling
  - different intervention design and measurement
  - models tailored to specific experiment

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- reproduce several prominent experiments in Kenya within a single environment
  - approximate sampling, intervention, measurement of each
  - unconditional cash transfers *(Egger et al., 2022; Miner, 2025)*
  - unemployment insurance *(Miner, 2025)*
  - information treatments *(Barnett-Howell et al., 2025)*

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  - unconditional cash transfers *(Egger et al., 2022; Miner, 2025)*
  - unemployment insurance *(Miner, 2025)*
  - information treatments *(Barnett-Howell et al., 2025)*
- use the estimated model to
  1. assess long-run effects of each intervention beyond the experimental horizon
  2. evaluate macro effects of scaling up each intervention nationally
  3. decompose the urban-rural income gap into contributions from different frictions

Preview: Model

heterogeneous agent general equilibrium Aiyagari model

- **households** choose consumption, savings, and location each period

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- solve for **stationary general equilibrium**

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  - unemployment insurance highly effective at increasing urbanization
- decomposing the urban–rural income gap:
  - information friction least important
  - financial friction explains much of the current migration, small part of the gap
  - residual migration costs explain most of the URPG

Experiments Background

Overview of Experiments

- several experiments conducted in Kenya, targeting different barriers to migration
- cash transfer:
  - *Egger et al. (2022)*
  - *Miner (2025)*
- unemployment insurance: *Miner (2025)*
- information provision: *Barnett-Howell et al. (2025)*

Cash Transfer: Egger et al. (2022)

Egger, Haushofer, Miguel, Niehaus, Walker (2022)

- **sample**: 10,500 poor households across 653 rural villages
  - randomly sampled from 653 villages within 1 county
- **intervention**: one-time huge unconditional cash transfer
  - 75% of median annual household income
  - 16% of annual village GDP
- **findings**:
  - expenditure $\uparrow$ 12%
  - no migration effect
  - large local spillovers: consumption $\uparrow$ similarly among treated and untreated
  - no change in local labor supply
  - local firms: sales $\uparrow$, wages $\uparrow$
- **requirements for model**:
  - financial frictions
  - mechanism for large local spillovers through demand channel (*WIP*)

Cash Transfer: Miner (2025)

Miner (2025)

- **sample**: 700 young male workers
  - randomly sampled from villages within 1 county
- **intervention**: one-time unconditional cash transfer similar to maximum UI benefits in UI arm
- **findings**:
  - migration rate $\uparrow$ 5 p.p.
  - no significant income change

Unemployment Insurance: Miner (2025)

Miner (2025)

- **sample**: 700 young male workers
  - randomly sampled from villages within 1 county
- **intervention**: offer daily UI in Nairobi
  - $\sim$ daily urban wage
  - up to 14 days
- **findings**:
  - migration rate $\uparrow$ 11 p.p.
  - no significant income change
- **requirements for model**:
  - unemployment risk

Information Provision: Barnett-Howell et al. (2025)

Barnett-Howell, Baseler, Ginn, Gordeev (2025)

- **sample**: 17,000 rural households
  - representatively sampled from 5 rural counties
- **intervention**: information about Nairobi income, labor market
- treatment arms:
  - **information**: individual delivery
  - **mentor**: individual + mentor pairing
  - **group**: delivery in village-wide meeting with group discussion

Information Provision: Barnett-Howell et al. (2025)

Barnett-Howell, Baseler, Ginn, Gordeev (2025)

- **findings**:
  - villagers underestimate conditional urban income by 44%
  - treatment $\rightarrow$ migration rate $\uparrow$ 2 p.p.
  - individual/mentor treatment $\rightarrow$ income $\uparrow$ 9%
  - group treatment $\rightarrow$ no income change
  - group treatment primarily engages experienced migrants, who have low marginal returns (because of low migration cost)
  - large economic spillovers within village
- **requirements for model**:
  - information friction
  - heterogeneous migration experience among villagers affecting migration cost

Model

Model Overview

Aiyagari-style dynamic heterogeneous agent model with migration

Regions

- Rural (R) and Urban (U)

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Frictions

- borrowing constraint: $a \geq \underline{a}$
- information friction:
  - rural households perceive $\tilde{w}_U = \gamma w_U$
- urban unemployment risk:
  - job separation λ_s
  - job finding λ_f
- migration costs: pay m_c when moving
- migration disutility:
  - the inexperienced discount urban consumption by m_u

Preferences and Experience

- CRRA utility over effective consumption:

$$u(c, l, x) = \frac{\tilde{c}^{1-\sigma} - 1}{1-\sigma}, \qquad \tilde{c} = \frac{c}{1 + m_u \cdot \mathbf{1}(l = U, x = 0)}$$

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  - inexperienced migrants in urban region discount consumption

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- m_u: consumption-equivalent disutility for inexperienced urban migrants
  - inexperienced migrants in urban region discount consumption
- urban experience $x \in \{0, 1\}$:
  - gain experience ($x' = 1$) with prob. π_x per period in urban, stay experienced
  - lose experience ($x' = 0$) with prob. π_{-x} per period in rural, stay inexperienced
  - as in *Lagakos, Mobarak, Waugh (2023)*

budget constraint: $c + a' + \mathbf{1}(l = R, l' = U) \cdot m_c = (1 + r)a + y(e, l, z_R, z_U)$

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EMPLOYMENT

- urban: stochastic employment e
  - $e = 1 \to 0$: job separation rate λ_s
  - $e = 0 \to 1$: job finding rate λ_f
- rural: always employed ($e = 1$)

Income, Productivity, and Employment

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- pay m_c when moving $R \rightarrow U$

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Bivariate AR(1) Productivity Process

$$\log z'_R = \rho_R \log z_R + \epsilon'_R$$
$$\log z'_U = \rho_U \log z_U + \epsilon'_U$$

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Assets

- save in one-period risk-free bond a'
  - return r
- borrowing constraint: $a' \geq \underline{a}$

Information Friction

- rural households perceive urban wage as $\tilde{w}_U = \gamma w_U$, where $\gamma \in (0, 1]$
  - $\gamma < 1$: rural households underestimate potential urban income
  - perceived income:

$$\tilde{y}(e, l, z_R, z_U, \gamma) = \begin{cases} w_R z_R & \text{if } l = R \\ \gamma w_U z_U & \text{if } l = U \text{ and } e = 1 \\ b & \text{if } l = U \text{ and } e = 0 \end{cases}$$

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  - as in *Barnett-Howell et al. (2025)*
- rural→urban migrants are surprised by true urban wage
- urban households don't anticipate forgetting the true urban wage
  - but revert to misperception if they return to rural

Recursive Formulation: Rural Household

rural household stays or migrates to urban:

$$V_R(a, z_R, z_U, x, \gamma) = \max\left\{V_{R\to R},\ V_{R\to U}\right\}$$

Stay in rural

$$V_{R\to R} = \max_{a' \geq \underline{a}} \quad u(c, R, x) + \beta\,\mathbb{E}[V_R(a', z_R', z_U', x', \gamma)]$$

$$c + a' = (1+r)a + \tilde{y}(1, R, z_R, z_U, \gamma)$$

Migrate to urban

$$V_{R\to U} = \max_{a' \geq \underline{a}} \quad u(c, R, x) + \beta\,\mathbb{E}[V_U(a', z_R', z_U', x', e', \gamma)]$$

$$c + a' + m_c = (1+r)a + \tilde{y}(1, R, z_R, z_U, \gamma)$$

Recursive Formulation: Urban Household

urban household stays or migrates to rural:

$$V_U(a, z_R, z_U, x, e, \gamma) = \max \left\{ V_{U\to U},\ V_{U\to R} \right\}$$

Stay in urban

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Migrate to rural

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Belief Dynamics of Urban Households

1. **Income beliefs:** while in U, households use V with $\gamma = 1$
   - do not anticipate forgetting the true urban wage after returning to R
   - otherwise everyone becomes well-informed in stationary equilibrium

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1. **Income beliefs:** while in U, households use V with $\gamma = 1$
   - do not anticipate forgetting the true urban wage after returning to R
   - otherwise everyone becomes well-informed in stationary equilibrium
2. **Migration cost beliefs:** while in U, households use V with $m_c = 0$
   - do not anticipate facing migration cost after returning to R for repeat $R \to U$ moves
   - allows the stationary equilibrium to hit URPG, urban share moments

Production

- production in each location l:

$$Y_l = A_l K_l^{\alpha} L_l^{1-\alpha}$$

- competitive input markets:

$$w_l = (1-\alpha) A_l (K_l / L_l)^{\alpha}$$

$$r + \delta = \alpha A_l (K_l / L_l)^{\alpha - 1}$$

- perfect capital mobility
  - r equalized across regions

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Equilibrium

1. household optimization given prices
   - rural households solve V_R with $\gamma < 1$
   - urban households solve V_U with $\gamma = 1$
2. firm optimization
3. market clearing:
   - labor: $L_l = \int z_l \, d\mu$
   - capital: $K_R + K_U = \int a \, d\mu$
4. stationary distribution

Estimation

Estimation

Assigned Parameters

t	period length
β	discount factor
σ	risk aversion
α	capital share
δ	depreciation
A_R	rural TFP
λ_s	job separation
λ_f	job finding
m_u	migration disutility
π_x	experience gain
π_{-x}	experience loss
γ	info friction
b	UI benefit
$\underline{a}$	borrowing limit

Estimated Parameters

A_U	urban TFP
ρ_R	rural persistence
ρ_U	urban persistence
$\sigma_{\epsilon,R}$	rural shock SD
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ρ_ϵ	shock correlation
m_c	migration cost

$t = 6$ months
length of one model period

- maximizes available data moments for estimation

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$\beta = 0.98$

discount factor

$$V_{l \to l'} = \max_{a' \geq \underline{a}} u(c, R, x) + \beta \, \mathbb{E}\left[V_{l'}(a', z'_R, z'_U, x', e', \gamma)\right]$$

- standard for 6-month period length (from 0.96 for annual)

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$\sigma = 2$

coefficient of risk aversion in CRRA utility

$$u(c, l, x) = \frac{\tilde{c}^{1-\sigma} - 1}{1 - \sigma}$$

- standard

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$\alpha = 0.64$

capital share in production

$$Y_l = A_l K_l^{\alpha} L_l^{1-\alpha}$$

- labor share: 0.36
- **data**: ILOSTAT

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$\delta = 0.051$

capital depreciation rate

$$r + \delta = \alpha A_l \left(K_l / L_l\right)^{\alpha - 1}$$

- standard for 6-month period length (from 0.10 for annual)

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b	UI benefit
$\underline{a}$	borrowing limit

Estimated Parameters

A_U	urban TFP
ρ_R	rural persistence
ρ_U	urban persistence
$\sigma_{\epsilon,R}$	rural shock SD
$\sigma_{\epsilon,U}$	urban shock SD
ρ_ϵ	shock correlation
m_c	migration cost

$A_R = 1$

rural total factor productivity

$$Y_R = A_R K_R^{\alpha} L_R^{1-\alpha}$$

- normalized

Estimation

Assigned Parameters	
t	period length
β	discount factor
σ	risk aversion
α	capital share
δ	depreciation
A_R	rural TFP
λ_s	job separation
λ_f	job finding
m_u	migration disutility
π_x	experience gain
π_{-x}	experience loss
γ	info friction
b	UI benefit
$\underline{a}$	borrowing limit
Estimated Parameters	
A_U	urban TFP
ρ_R	rural persistence
ρ_U	urban persistence
$\sigma_{\epsilon,R}$	rural shock SD
$\sigma_{\epsilon,U}$	urban shock SD
ρ_ϵ	shock correlation
m_c	migration cost

$\lambda_s = 0.06$
urban job separation rate

$$\Pr(e' = 0 \mid e = 1, l = U) = \lambda_s, \qquad \Pr(e' = 1 \mid e = 0, l = U) = \lambda_f$$

- **target moment**: 6-month unemployment hazard rate for urban migrants
- **data**: survey in *Barnett-Howell et al. (2025)*

Estimation

Assigned Parameters

t	period length
β	discount factor
σ	risk aversion
α	capital share
δ	depreciation
A_R	rural TFP
λ_s	job separation
λ_f	job finding
m_u	migration disutility
π_x	experience gain
π_{-x}	experience loss
γ	info friction
b	UI benefit
$\underline{a}$	borrowing limit

Estimated Parameters

A_U	urban TFP
ρ_R	rural persistence
ρ_U	urban persistence
$\sigma_{\epsilon,R}$	rural shock SD
$\sigma_{\epsilon,U}$	urban shock SD
ρ_ϵ	shock correlation
m_c	migration cost

$\lambda_f = 0.42$

urban job finding rate

$$\Pr(e' = 0 \mid e = 1, l = U) = \lambda_s, \qquad \Pr(e' = 1 \mid e = 0, l = U) = \lambda_f$$

- **target moment**: 6-month employment hazard rate for urban migrants
- **data**: survey in *Barnett-Howell et al. (2025)*

Estimation

Assigned Parameters

t	period length
β	discount factor
σ	risk aversion
α	capital share
δ	depreciation
A_R	rural TFP
λ_s	job separation
λ_f	job finding
m_u	migration disutility
π_x	experience gain
π_{-x}	experience loss
γ	info friction
b	UI benefit
$\underline{a}$	borrowing limit

Estimated Parameters

A_U	urban TFP
ρ_R	rural persistence
ρ_U	urban persistence
$\sigma_{\epsilon,R}$	rural shock SD
$\sigma_{\epsilon,U}$	urban shock SD
ρ_ϵ	shock correlation
m_c	migration cost

$m_u = 0.34$

consumption-equivalent utility cost of migration

$$\tilde{c} = \frac{c}{1 + m_u \cdot \mathbf{1}(l = U, x = 0)}$$

- **target moment**: share of income inexperienced migrants are willing to give up to move back to rural
  - survey asked migrants what rural income would make them willing to move back
- **data**: survey in *Barnett-Howell et al. (2025)*

Estimation

Assigned Parameters

t	period length
β	discount factor
σ	risk aversion
α	capital share
δ	depreciation
A_R	rural TFP
λ_s	job separation
λ_f	job finding
m_u	migration disutility
π_x	experience gain
π_{-x}	experience loss
γ	info friction
b	UI benefit
$\underline{a}$	borrowing limit

Estimated Parameters

A_U	urban TFP
ρ_R	rural persistence
ρ_U	urban persistence
$\sigma_{\epsilon,R}$	rural shock SD
$\sigma_{\epsilon,U}$	urban shock SD
ρ_ϵ	shock correlation
m_c	migration cost

$\pi_x = 0.45$

urban experience gaining rate

$$\Pr(x' = 1 \mid l = U, x = 0) = \pi_x, \qquad \Pr(x' = 0 \mid l = R, x = 1) = \pi_{-x}$$

- **target moment**: among rural households with rural preference, what share switches to urban preference within 6 months after migrating
  - location preference: require higher income to move to the other location
- **data**: survey in *Barnett-Howell et al. (2025)*

Estimation

Assigned Parameters

t	period length
β	discount factor
σ	risk aversion
α	capital share
δ	depreciation
A_R	rural TFP
λ_s	job separation
λ_f	job finding
m_u	migration disutility
π_x	experience gain
π_{-x}	experience loss
γ	info friction
b	UI benefit
$\underline{a}$	borrowing limit

Estimated Parameters

A_U	urban TFP
ρ_R	rural persistence
ρ_U	urban persistence
$\sigma_{\epsilon,R}$	rural shock SD
$\sigma_{\epsilon,U}$	urban shock SD
ρ_ϵ	shock correlation
m_c	migration cost

$\pi_{-x} = 0.68$

urban experience losing rate

$$\Pr(x' = 1 \mid l = U, x = 0) = \pi_x, \qquad \Pr(x' = 0 \mid l = R, x = 1) = \pi_{-x}$$

- **target moment**: among urban households with urban preference, what share switches to rural preference within 6 months after migrating
  - location preference: require higher income to move to the other location
- **data**: survey in *Barnett-Howell et al. (2025)*

Estimation

Assigned Parameters

t	period length
β	discount factor
σ	risk aversion
α	capital share
δ	depreciation
A_R	rural TFP
λ_s	job separation
λ_f	job finding
m_u	migration disutility
π_x	experience gain
π_{-x}	experience loss
γ	info friction
b	UI benefit
$\underline{a}$	borrowing limit

Estimated Parameters

A_U	urban TFP
ρ_R	rural persistence
ρ_U	urban persistence
$\sigma_{\epsilon,R}$	rural shock SD
$\sigma_{\epsilon,U}$	urban shock SD
ρ_ϵ	shock correlation
m_c	migration cost

$\gamma = 0.56$

understatement of urban income by rural households

$$\tilde{w}_U = \gamma w_U$$

- **target moment**: avg expected Nairobi income by rural households as a share of actual Nairobi income, conditional on demographics
- **data**: *Barnett-Howell et al. (2025)*

Estimation

Assigned Parameters

t	period length
β	discount factor
σ	risk aversion
α	capital share
δ	depreciation
A_R	rural TFP
λ_s	job separation
λ_f	job finding
m_u	migration disutility
π_x	experience gain
π_{-x}	experience loss
γ	info friction
b	UI benefit
$\underline{a}$	borrowing limit

Estimated Parameters

A_U	urban TFP
ρ_R	rural persistence
ρ_U	urban persistence
$\sigma_{\epsilon,R}$	rural shock SD
$\sigma_{\epsilon,U}$	urban shock SD
ρ_ϵ	shock correlation
m_c	migration cost

$b = 0$

urban unemployment benefit

$$y(e, U, z_R, z_U) = \begin{cases} w_U z_U & e = 1 \\ b & e = 0 \end{cases}$$

- no formal unemployment insurance in Kenya at baseline

Estimation

Assigned Parameters

t	period length
β	discount factor
σ	risk aversion
α	capital share
δ	depreciation
A_R	rural TFP
λ_s	job separation
λ_f	job finding
m_u	migration disutility
π_x	experience gain
π_{-x}	experience loss
γ	info friction
b	UI benefit
$\underline{a}$	borrowing limit

Estimated Parameters

A_U	urban TFP
ρ_R	rural persistence
ρ_U	urban persistence
$\sigma_{\epsilon,R}$	rural shock SD
$\sigma_{\epsilon,U}$	urban shock SD
ρ_ϵ	shock correlation
m_c	migration cost

$\underline{a} = 0$

borrowing limit

$$c + a' + \mathbf{1}(l = R, l' = U) \cdot m_c = (1 + r)a + y(e, l, z_R, z_U), \qquad a' \geq \underline{a}$$

- no borrowing allowed at baseline

Estimation

Assigned Parameters

t	period length
β	discount factor
σ	risk aversion
α	capital share
δ	depreciation
A_R	rural TFP
λ_s	job separation
λ_f	job finding
m_u	migration disutility
π_x	experience gain
π_{-x}	experience loss
γ	info friction
b	UI benefit
$\underline{a}$	borrowing limit

Estimated Parameters

A_U	urban TFP
ρ_R	rural persistence
ρ_U	urban persistence
$\sigma_{\epsilon,R}$	rural shock SD
$\sigma_{\epsilon,U}$	urban shock SD
ρ_ϵ	shock correlation
m_c	migration cost

$A_U = 1.77$

urban total factor productivity

$$Y_U = A_U K_U^{\alpha} L_U^{1-\alpha}$$

- **target moment**: urban–rural ratio of median incomes
- **data**: Kenya Integrated Household Budget Survey (KIHBS)

Estimation

Assigned Parameters

t	period length
β	discount factor
σ	risk aversion
α	capital share
δ	depreciation
A_R	rural TFP
λ_s	job separation
λ_f	job finding
m_u	migration disutility
π_x	experience gain
π_{-x}	experience loss
γ	info friction
b	UI benefit
$\underline{a}$	borrowing limit

Estimated Parameters

A_U	urban TFP
ρ_R	rural persistence
ρ_U	urban persistence
$\sigma_{\epsilon,R}$	rural shock SD
$\sigma_{\epsilon,U}$	urban shock SD
ρ_ϵ	shock correlation
m_c	migration cost

$\rho_R = 0.64$

persistence of rural productivity

$$\log z_R' = \rho_R \log z_R + \epsilon_R'$$
$$\log z_U' = \rho_U \log z_U + \epsilon_U'$$
$$\text{SD}(\epsilon_R') = \sigma_{\epsilon,R}, \quad \text{SD}(\epsilon_U') = \sigma_{\epsilon,U}$$
$$\text{Corr}(\epsilon_R', \epsilon_U') = \rho_\epsilon$$

- **target moment**: standard deviation of log rural income
- **data**: Kenya Integrated Household Budget Survey (KIHBS)

Estimation

Assigned Parameters

t	period length
β	discount factor
σ	risk aversion
α	capital share
δ	depreciation
A_R	rural TFP
λ_s	job separation
λ_f	job finding
m_u	migration disutility
π_x	experience gain
π_{-x}	experience loss
γ	info friction
b	UI benefit
$\underline{a}$	borrowing limit

Estimated Parameters

A_U	urban TFP
ρ_R	rural persistence
ρ_U	urban persistence
$\sigma_{\epsilon,R}$	rural shock SD
$\sigma_{\epsilon,U}$	urban shock SD
ρ_ϵ	shock correlation
m_c	migration cost

$\rho_U = 0.78$

persistence of urban productivity

$$\log z_R' = \rho_R \log z_R + \epsilon_R'$$

$$\log z_U' = \rho_U \log z_U + \epsilon_U'$$

$$\text{SD}(\epsilon_R') = \sigma_{\epsilon,R}, \quad \text{SD}(\epsilon_U') = \sigma_{\epsilon,U}$$

$$\text{Corr}(\epsilon_R', \epsilon_U') = \rho_\epsilon$$

- **target moment**: standard deviation of log urban income
- **data**: Kenya Integrated Household Budget Survey (KIHBS)

Estimation

Assigned Parameters

t	period length
β	discount factor
σ	risk aversion
α	capital share
δ	depreciation
A_R	rural TFP
λ_s	job separation
λ_f	job finding
m_u	migration disutility
π_x	experience gain
π_{-x}	experience loss
γ	info friction
b	UI benefit
$\underline{a}$	borrowing limit

Estimated Parameters

A_U	urban TFP
ρ_R	rural persistence
ρ_U	urban persistence
$\sigma_{\epsilon,R}$	rural shock SD
$\sigma_{\epsilon,U}$	urban shock SD
ρ_ϵ	shock correlation
m_c	migration cost

$\sigma_{\epsilon,R} = 1.16$

volatility of rural productivity shocks

$$\log z'_R = \rho_R \log z_R + \epsilon'_R$$
$$\log z'_U = \rho_U \log z_U + \epsilon'_U$$
$$\text{SD}(\epsilon'_R) = \sigma_{\epsilon,R}, \quad \text{SD}(\epsilon'_U) = \sigma_{\epsilon,U}$$
$$\text{Corr}(\epsilon'_R, \epsilon'_U) = \rho_\epsilon$$

- **target moment**: standard deviation of residual rural log income after controlling for persistence
- **data**: survey in *Barnett-Howell et al. (2025)*

Estimation

Assigned Parameters

t	period length
β	discount factor
σ	risk aversion
α	capital share
δ	depreciation
A_R	rural TFP
λ_s	job separation
λ_f	job finding
m_u	migration disutility
π_x	experience gain
π_{-x}	experience loss
γ	info friction
b	UI benefit
$\underline{a}$	borrowing limit

Estimated Parameters

A_U	urban TFP
ρ_R	rural persistence
ρ_U	urban persistence
$\sigma_{\epsilon,R}$	rural shock SD
$\sigma_{\epsilon,U}$	urban shock SD
ρ_ϵ	shock correlation
m_c	migration cost

$\sigma_{\epsilon,U} = 1.07$

volatility of urban productivity shocks

$$\log z_R' = \rho_R \log z_R + \epsilon_R'$$
$$\log z_U' = \rho_U \log z_U + \epsilon_U'$$
$$\mathrm{SD}(\epsilon_R') = \sigma_{\epsilon,R}, \quad \mathrm{SD}(\epsilon_U') = \sigma_{\epsilon,U}$$
$$\mathrm{Corr}(\epsilon_R', \epsilon_U') = \rho_\epsilon$$

- **target moment**: standard deviation of residual urban log income after controlling for persistence
- **data**: survey in *Barnett-Howell et al. (2025)*

Estimation

Assigned Parameters

t	period length
β	discount factor
σ	risk aversion
α	capital share
δ	depreciation
A_R	rural TFP
λ_s	job separation
λ_f	job finding
m_u	migration disutility
π_x	experience gain
π_{-x}	experience loss
γ	info friction
b	UI benefit
$\underline{a}$	borrowing limit

Estimated Parameters

A_U	urban TFP
ρ_R	rural persistence
ρ_U	urban persistence
$\sigma_{\epsilon,R}$	rural shock SD
$\sigma_{\epsilon,U}$	urban shock SD
ρ_ϵ	shock correlation
m_c	migration cost

$\rho_\epsilon = 0.03$

cross-location correlation of productivity shocks within a household.

$$\log z'_R = \rho_R \log z_R + \epsilon'_R$$
$$\log z'_U = \rho_U \log z_U + \epsilon'_U$$
$$\mathrm{SD}(\epsilon'_R) = \sigma_{\epsilon,R}, \quad \mathrm{SD}(\epsilon'_U) = \sigma_{\epsilon,U}$$
$$\mathrm{Corr}(\epsilon'_R, \epsilon'_U) = \rho_\epsilon$$

- **target moment**: slope from regressing urban log income on rural log income for similar households (clustered by observables)
- **model moment**: slope from regressing potential urban log income on potential rural log income within household
- **data**: *Barnett-Howell et al. (2025)*

Estimation

Assigned Parameters

t	period length
β	discount factor
σ	risk aversion
α	capital share
δ	depreciation
A_R	rural TFP
λ_s	job separation
λ_f	job finding
m_u	migration disutility
π_x	experience gain
π_{-x}	experience loss
γ	info friction
b	UI benefit
$\underline{a}$	borrowing limit

Estimated Parameters

A_U	urban TFP
ρ_R	rural persistence
ρ_U	urban persistence
$\sigma_{\epsilon,R}$	rural shock SD
$\sigma_{\epsilon,U}$	urban shock SD
ρ_ϵ	shock correlation
m_c	migration cost

$m_c = 1721$

monetary migration cost

$$c + a' + \mathbf{1}(l = R, l' = U)\, m_c = (1 + r)a + y(e, l, z_R, z_U)$$

- **target moment**: share of Kenyan population living in cities
- **model moment**: share of households in urban region
- **data**: Kenya National Bureau of Statistics

Estimated Parameters and Target Moments

Parameter	Value	Target moment	Moment value	
			Data	Model
A_U	1.77	Urban–rural ratio of median incomes	5.15	4.95
ρ_R	0.64	Standard deviation of log rural income	1.49	1.55
ρ_U	0.78	Standard deviation of log urban income	1.57	1.60
$\sigma_{\epsilon,R}$	1.16	Standard deviation of residual rural log income	1.14	1.19
$\sigma_{\epsilon,U}$	1.07	Standard deviation of residual urban log income	1.05	1.10
ρ_ϵ	0.03	Slope of urban vs rural log incomes	0.065	0.068
m_c	1721	Urban population share	0.320	0.328

Reproducing Experiments

Cash Transfer: Egger et al. (2022)

Sampling

RCT

- 1 rural Kenyan county
- half of villages treated
- eligibility: thatched roof (1/3 of households)

Model

- eligibility: poorest 1/3 of rural households by assets
- randomly treat half

Cash Transfer: Egger et al. (2022)

Sampling

RCT

- 1 rural Kenyan county
- half of villages treated
- eligibility: thatched roof (1/3 of households)

Model

- eligibility: poorest 1/3 of rural households by assets
- randomly treat half

Treatment

RCT

- one-time unconditional transfer
- 75% of mean annual expenditure

Model

- one-time asset transfer
- 75% of mean rural annual consumption

Cash Transfer: Egger et al. (2022)

Sampling

RCT

- 1 rural Kenyan county
- half of villages treated
- eligibility: thatched roof (1/3 of households)

Model

- eligibility: poorest 1/3 of rural households by assets
- randomly treat half

Treatment

RCT

- one-time unconditional transfer
- 75% of mean annual expenditure

Model

- one-time asset transfer
- 75% of mean rural annual consumption

Measurement

RCT

- endline: 19 months after transfer
- migration and household expenditure, treatment vs control

Model

- endline: 3 periods (18 months)
- migration: urban residence at endline
- consumption

Cash Transfer: Egger et al. (2022) Results

	Migration Rate Change		Consumption Change	
treatment arm	RCT	Model	RCT	Model
cash transfer	0.0 pp	0.0 pp	11.6%	13.7%

- model matches negligible short-run migration response, magnitude of consumption response

Unemployment Insurance and Cash Transfer: Miner (2025)

Sampling

RCT

- male rural workers aged 18–30 in 1 rural county
- most had prior migration experience
- sampled 11.7% of rural households per village

Model

- sample 11.7% of rural households
- prioritize experienced rural households

Unemployment Insurance and Cash Transfer: Miner (2025)

Sampling

RCT

- male rural workers aged 18–30 in 1 rural county
- most had prior migration experience
- sampled 11.7% of rural households per village

Model

- sample 11.7% of rural households
- prioritize experienced rural households

Treatment

RCT

- UI: up to 15 days at average urban wage (60 USD)
- cash: one-time 50 USD transfer

Model

- UI: b matching 15 days of avg urban wage
- cash: asset transfer, 50/60 of UI amount

Unemployment Insurance and Cash Transfer: Miner (2025)

Sampling	Treatment	Measurement
RCT • male rural workers aged 18–30 in 1 rural county • most had prior migration experience • sampled 11.7% of rural households per village	**RCT** • UI: up to 15 days at average urban wage (60 USD) • cash: one-time 50 USD transfer	**RCT** • endline: about 6 months after treatment • Nairobi residence and monthly income
Model • sample 11.7% of rural households • prioritize experienced rural households	**Model** • UI: b matching 15 days of avg urban wage • cash: asset transfer, 50/60 of UI amount	**Model** • endline: 1 period (6 months) • urban residence and income changes relative to rural baseline income

Unemployment Insurance and Cash Transfer: Miner (2025) Results

	Migration Rate Change		Income Change	
treatment arm	RCT	Model	RCT	Model
unemployment insurance	10.9 pp	10.8 pp	0.0%	120.5%

- UI: model matches migration response, vastly overpredicts income response

Unemployment Insurance and Cash Transfer: Miner (2025) Results

	Migration Rate Change		Income Change	
treatment arm	RCT	Model	RCT	Model
unemployment insurance	10.9 pp	10.8 pp	0.0%	120.5%
cash transfer	4.5 pp	3.0 pp	0.0%	9.1%

- UI: model matches migration response, vastly overpredicts income response
- cash: model matches migration response, overpredicts income response
  - migration response exceeds that in *Egger et al. (2022)* due to sample of experienced migrants

Information Provision: Barnett-Howell et al. (2025)

Sampling

RCT

- 17,000 rural households in 5 counties
- 30% of households within sampled villages treated

Model

- randomly treat 30% of rural households

Information Provision: Barnett-Howell et al. (2025)

Sampling	Treatment

RCT

- 17,000 rural households in 5 counties
- 30% of households within sampled villages treated

Model

- randomly treat 30% of rural households

RCT

- information: beliefs $\uparrow$ 4%
- mentor: beliefs $\uparrow$ 10%
- group: beliefs $\uparrow$ 12%

Model

- information: $\gamma_{treated} = 1.04\gamma$
- mentor: $\gamma_{treated} = 1.10\gamma$
- group: experienced treated households set to $\gamma = 1$; inexperienced γ chosen to match average belief gain

Information Provision: Barnett-Howell et al. (2025)

Sampling

RCT

- 17,000 rural households in 5 counties
- 30% of households within sampled villages treated

Model

- randomly treat 30% of rural households

Treatment

RCT

- information: beliefs $\uparrow$ 4%
- mentor: beliefs $\uparrow$ 10%
- group: beliefs $\uparrow$ 12%

Model

- information: $\gamma_{treated} = 1.04\gamma$
- mentor: $\gamma_{treated} = 1.10\gamma$
- group: experienced treated households set to $\gamma = 1$; inexperienced γ chosen to match average belief gain

Measurement

RCT

- endline: about 16 months after treatment
- any Nairobi migrant and household income

Model

- endline: 3 periods (18 months)
- urban residence and income changes relative to rural baseline income

Information Provision: Results

	Migration Rate Change		Income Change	
treatment arm	RCT	Model	RCT	Model
information	1.8 pp	0.8 pp	6.7%	3.1%

- information: model underpredicts migration and income responses

Information Provision: Results

treatment arm	Migration Rate Change		Income Change	
	RCT	Model	RCT	Model
information	1.8 pp	0.8 pp	6.7%	3.1%
mentor	2.4 pp	2.4 pp	7.7%	7.1%

- information: model underpredicts migration and income responses
- mentor: model matches responses well
  - information arm has to be worse because of a smaller impact on beliefs

Information Provision: Results

	Migration Rate Change		Income Change	
treatment arm	RCT	Model	RCT	Model
information	1.8 pp	0.8 pp	6.7%	3.1%
mentor	2.4 pp	2.4 pp	7.7%	7.1%
group	2.2 pp	2.4 pp	0.0%	6.9%

- information: model underpredicts migration and income responses
- mentor: model matches responses well
  - information arm has to be worse because of a smaller impact on beliefs
- group: model matches migration response, overpredicts income response
  - income response smaller than mentor due to selection of experienced, low-return migrants
  - but difference between mentor and group is tiny in the model

Quantitative Results

Quantitative Exercises

- track **long-run** treatment effects beyond each RCT endline

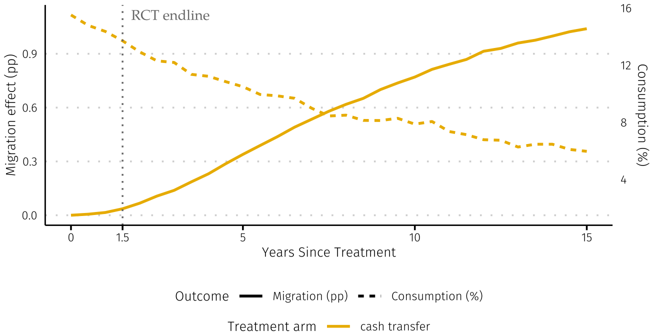
Quantitative Exercises

- track **long-run** treatment effects beyond each RCT endline
- roll out interventions **nationally** and re-solve general equilibrium

Quantitative Exercises

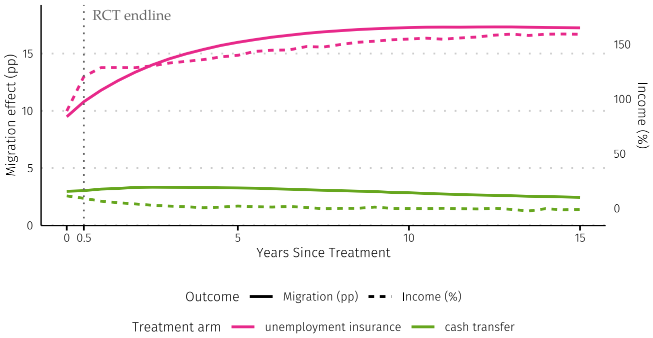
- track **long-run** treatment effects beyond each RCT endline
- roll out interventions **nationally** and re-solve general equilibrium
- shut down individual frictions to **decompose** the urban–rural income gap

Long-Run Effects: Cash Transfer, Egger et al. (2022)



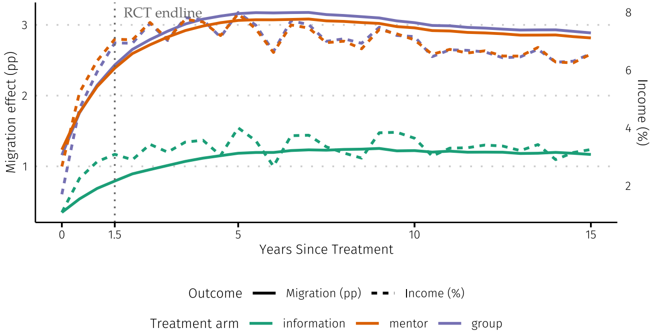
- consumption effects decay slowly as the transfer is spent
- migration is negligible at endline, but builds to 1 pp after 15 years
  - as more and more rural households experience shocks making migration attractive

Long-Run Effects: UI and Cash Transfer, Miner (2025)



- UI: 2/3 of effects realized by endline, continue to build
  - persistence partly reflects the model treatment: UI is available in every 6-month period
- cash: effects mostly realized by endline, then decay

Long-Run Effects: Information Provision



- 3/4 of effects realized by endline, continue to build
- gains continue growing for a few years as favorable shocks make migration attractive
- treatment-control gaps later shrink: beliefs are identical when in the city

National Rollout

Scenario	Urban Share	URPG
Baseline	32.8%	4.945

National Rollout

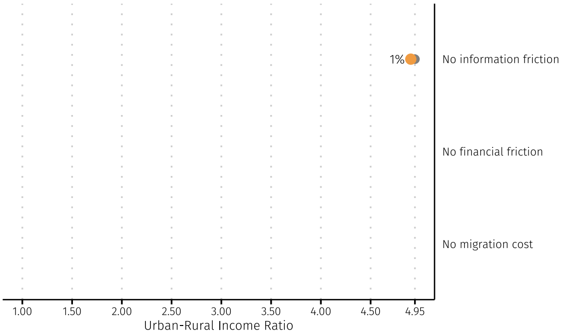
Scenario	Urban Share	URPG
Baseline	32.8%	4.945
Barnett-Howell et al.: Individual	33.5%	4.940

Scenario	Urban Share	URPG
Baseline	32.8%	4.945
Barnett-Howell et al.: Individual	33.5%	4.940
Barnett-Howell et al.: Mentor	35.6%	4.928

National Rollout

Scenario	Urban Share	URPG
Baseline	32.8%	4.945
Barnett-Howell et al.: Individual	33.5%	4.940
Barnett-Howell et al.: Mentor	35.6%	4.928
Miner: Unemployment insurance	58.1%	5.046

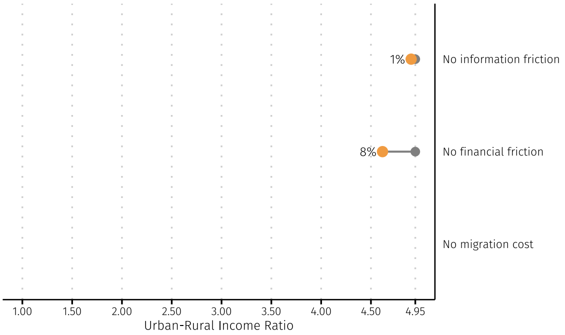
Urban–Rural Income Gap Decomposition



Scenario	Urban Share	URPG
Baseline	32.8%	4.945
No information friction	32.9%	4.904

- information explains little of URPG

Urban–Rural Income Gap Decomposition



Scenario	Urban Share	URPG
Baseline	32.8%	4.945
No information friction	32.9%	4.904
No financial friction	1.2%	4.616

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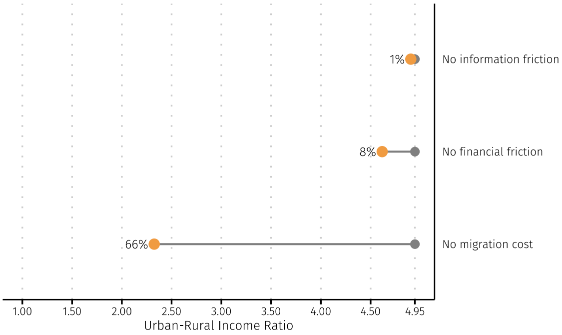

Scenario	Urban Share	URPG
Baseline	32.8%	4.945
No information friction	32.9%	4.904
No financial friction	1.2%	4.616
No migration cost	72.9%	2.325

- information explains little of URPG
- removing borrowing constraint lowers the gap but collapses urban share
  - suggests that much of current migration is a form of self-insurance in the absence of formal credit markets
- residual migration costs are the dominant barrier in the model

Conclusion

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- **future steps**:
  - refine experiment mapping, estimation, simulation
  - alternative experiment mapping: calibrate treatment intensity to match treatment effects
  - non-linearities in firm production (slack) to reproduce local economic spillovers
  - cost-effectiveness comparison